

Switched Capacitor Circuits

Exercise 1: Sample and hold circuit

Simulate simple sample and hold circuit as shown in Fig. ?? . Apply a sine wave input of 1KHz with $200mV_{pp}$ over a dc voltage of 0.1V. Apply clock of 10 KHz with high and low values of 1.8 V and 0 V respectively. Plot the output on node 2 and observe the effect of charge injection and clock feedthrough on the output.

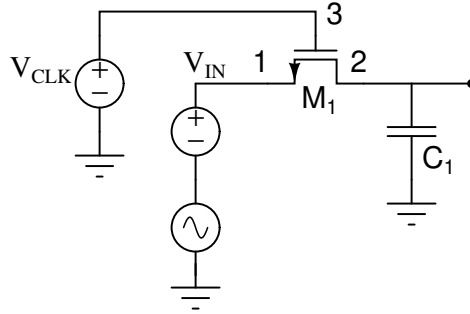


Figure 1: Sample and hold circuit

Exercise 2: Sample and hold circuit with bottom plate sampling

Simulate the sample and hold circuit with bottom plate sampling. Apply clk1 signal with the delay of $10 \mu s$ with respect to clk2.

- Plot the voltage on node 2 and comment on the effect of charge injection and clock feedthrough effects.
- Repeat the (a) with 0 delay between the clock signals.

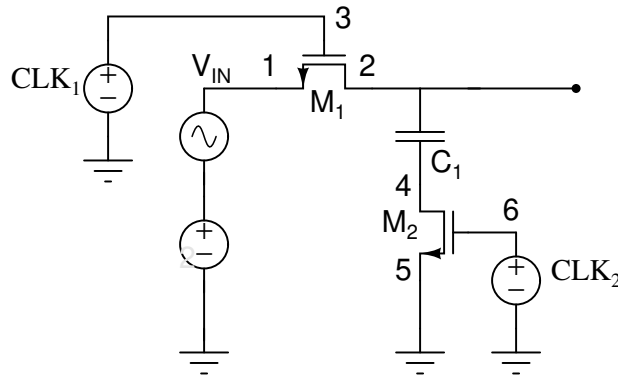


Figure 2: Sample and hold circuit

Exercise 3: Switched Capacitor Resistor

Simulate the switched capacitor resistor shown in Fig. 3. Apply the voltage $V_1 = 0.6V$, $V_2 = 0.4V$, with non overlapping clock pulses.

- Measure the average current through voltage V_{1a} and evaluate the resistance emulated by the above circuit using $R = T/C_1$.

- b) Evaluate the current through R when connected between the voltages V_1 and V_2 (as shown in Fig. 4).
c) Repeat (a),(b) for different values of C_1 and T .

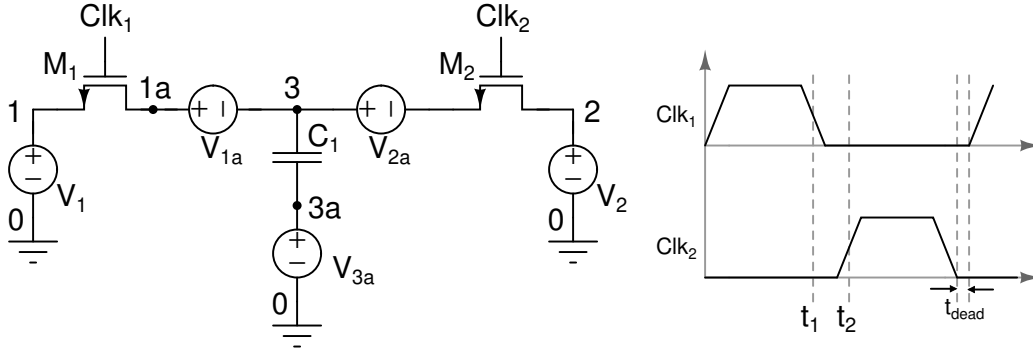


Figure 3: MOS Switched Capacitor resistor

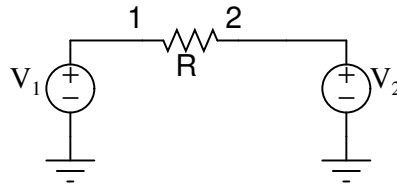


Figure 4: Resistor R connected between V_1 and V_2 .

Exercise 4: Switched Capacitor Integrator

Simulate the switched capacitor integrator as shown in Fig. ???. Apply voltage $V_1 = 0.6V$ with non-overlapping clock signals Clk_1 and Clk_2 .

- (a) Use the ideal OPAMP for your simulation. Plot the voltage on node 4 and observe the integration operation.
(b) Repeat (a) with previously designed OPAMP.

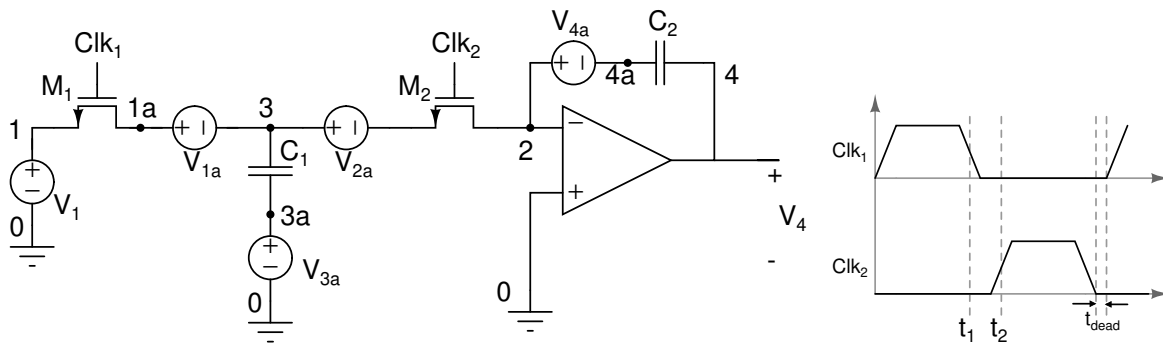


Figure 5: MOS Switched Capacitor Integrator